

Syed Nazmus Sakib

Research Assistant, Cortex AI Lab | B.Sc. Student, University of Dhaka

 Portfolio |  Scholar |  GitHub |  LinkedIn |  Mail

RESEARCH INTERESTS

- **Agentic and multimodal AI:** how LLM agents and vision–language models investigate, reason and act under uncertainty; evaluating agent safety, robustness and multi-agent behavior beyond benchmark accuracy.
- **Reinforcement learning and physical intelligence:** how representations, rewards and evaluation shape learning; RL for LLM post-training (RLVR), goal-conditioned RL and robot learning for long-horizon manipulation.

EDUCATION

B.Sc. in Robotics and Mechatronics Engineering

University of Dhaka

Sep 2022 – Oct 2026 (expected)

Dhaka, Bangladesh

RESEARCH EXPERIENCE

Research Assistant

CortexAI Lab, University of Dhaka

Mar 2025 – Present

Dhaka, Bangladesh

- First-authored research on multimodal reasoning and autonomous-agent safety, including papers accepted at NeurIPS 2026, Findings of ACL 2026 and two NeurIPS 2026 workshops, and a dataset paper in Scientific Data.
- Built large-scale evaluation resources, including PlantInquiryVQA (138K QA pairs), PlantExpertVQA (193K QA pairs) and agentic moderation benchmarks.
- Trained MemeAgent, a 2B-parameter verifier reaching 87.8% in-distribution auditing accuracy and outperforming zero-shot frontier baselines.

Research Intern

Data and Design Lab, CARS, University of Dhaka

Nov 2024 – Present

Dhaka, Bangladesh

- Analyzed large-scale billing and distribution data for power-quality assessment in an industry-focused research project on power supply around Dhaka City.
- Developed machine learning pipelines for failure prediction, anomaly detection and load/demand forecasting from power-quality meter data.
- Contributed to an electricity-domain conversational assistant that answers billing, outage and power-quality queries in natural language.

R&D Engineer

Tech Topia

Nov 2023 – Jul 2025

Dhaka, Bangladesh

- Designed robotics and IoT automation systems using Arduino and ESP32, including hardware–software integration for industrial applications.

SELECTED PUBLICATIONS

MemeEconomy: Do LLM Agents Trade Ethics for Survival?

Syed Nazmus Sakib, Nafiul Haque, Ahnaf Tahmid Manan, M. M. Mahabub Morshed, Shifat E. Arman

Accepted, *NeurIPS 2026*

Multimodal multi-agent market simulation across 300 events; develops a compact verifier for auditing harmful agent behavior under survival pressure.

Thinking Like a Botanist: Challenging Multimodal Language Models with Intent-Driven Chain-of-Inquiry

Syed Nazmus Sakib, Nafiul Haque, Shahrear Bin Amin, H. M. Abdullah, M. M. Hasan, Mohammad Zabed Hossain, Shifat E. Arman

Accepted, *Findings of ACL 2026*

[ACL Anthology] [arXiv]

Introduces PlantInquiryVQA with 24,964 expert-curated images and 138,078 QA pairs; structured inquiry improves diagnostic correctness and reduces hallucination in vision–language models.

Unlearnable, or Unmeasured? On the Reliability of Difficulty Labels in RLVR

Chandak Chakma, Syed Nazmus Sakib, Nafiul Haque, Shifat E. Arman

Accepted, *NeurIPS 2026 Workshop on Transitioning from Pre-Training to Post-Training*

[arXiv] [project]

Shows that “unlearnable” prompts in RL with verifiable rewards do improve, at about one third of the learnable rate, and that the difficulty labels defining them are far less reproducible than assumed; gives a sampling framework for the evaluation budget needed.

The Surface You Test Is Not the Surface That Breaks

Syed Nazmus Sakib, Nafiul Haque, Shahrear Bin Amin, Shifat E. Arman

Accepted, NeurIPS 2026 Workshop on Agents in the Wild: Safety, Security, and Beyond [arXiv] [project] [code]

Shows that prompt-injection vulnerability is a model-by-surface interaction; a surface-adaptive attacker improves attack success across 13 language models.

Do Better Goal Representations Improve Goal-Conditioned Reinforcement Learning?

Syed Nazmus Sakib, Abdul Monaf Chowdhury, Nafiul Haque, Shifat E. Arman, Md Mehedi Hasan

Preprint [arXiv] [project]

Finds that goal-representation quality barely affects offline goal-conditioned RL on OGBench navigation, while the state pathway is the bottleneck; random Fourier positional encodings of the state more than double success on the hardest tasks.

Do Web Agents Investigate Before They Decide?

Syed Nazmus Sakib, Nafiul Haque, Tapodhir Karmakar Taton, Shahrear Bin Amin, Shifat E. Arman

Preprint [arXiv]

Benchmark of 750 multi-step tasks isolating investigative competence across eight LLM agents; reveals navigation–discovery gaps, contradiction failures and investigative hallucination.

Complete publication list: Google Scholar.

ENGINEERING PROJECTS

Long-Horizon Mobile Manipulation and Robust Skill Composition

2026 – Present

Independent research using ManiSkill / MS-HAB, PyTorch and simulation-based evaluation

- Reproduced and evaluated a long-horizon TidyHouse pipeline using 21 pretrained Pick, Place and Navigate policies, with strict success, reachability and failure-stage diagnostics.
- Designed controlled handoff experiments showing that robot-state mismatch, rather than object-state mismatch, dominated chained Pick→Place degradation; localized major physical failures to base/torso contacts and force-limit trips.
- Improved full-chain success from 0.7% to 2.2% by tuning global base damping and force limits, validated over three seeds with confidence intervals, per-skill analysis and representative rollout videos.

Mobile Differential-Drive Robot

2025

ROS 2, Gazebo, Xacro, URDF and autonomous exploration

GitHub

- Built a complete ROS 2 simulation package covering robot modeling, differential-drive control, sensor integration and autonomous exploration in Gazebo.

Hexapod Robot

2024

C++, CMake, inverse kinematics and gait generation

GitHub

- Implemented inverse-kinematics-based leg control, coordinated gait generation and motion planning for a six-legged robotic platform.

TECHNICAL SKILLS

Programming	Python, C++, SQL, JavaScript, TypeScript
Agentic AI	LLM agents, tool use, multi-agent simulation, prompt-injection robustness, benchmark design
Robot Learning	PyTorch, Gymnasium, ManiSkill / MS-HAB, reinforcement learning, policy evaluation
Simulation & Robotics	Isaac Sim, Isaac Lab, MuJoCo, ROS 2, Gazebo, URDF, Xacro, kinematics, motion planning
Perception & ML	OpenCV, Transformers, vision–language models, TensorFlow, Scikit-learn
Hardware & Engineering	Arduino, ESP32, Autodesk Fusion 360, Docker, Git, Linux, MLflow

HONORS & AWARDS

Global Nominee NASA Space Apps Challenge	2024
Runner-up DU AI Challenge, University of Dhaka	2025
Runner-up KUET Datathon, Machine Learning Competition	2025
Runner-up Technocrats V2 Hackathon, IUBAT	2024
Regional Champion National High School Programming Contest (NHSPC)	2019
Kaggle Expert Multiple podium finishes in machine learning competitions	Ongoing